We aim to find a set of functions $\{f_1, f_2, ..., f_T\}$ that minimize:

$$\mathcal{L}(\phi) = \sum_{i=1}^{n} L(y_i, \hat{y}_i) + \sum_k \Omega(f_k) \tag{8.5}$$

where $\Omega(f) = \gamma T + \frac{1}{2}||w||^2$. T is the number of leaves and w are the leaf weights (value of lead node).

Since we solve this additively, at step t, we seek f_t to minimize:

$$\mathcal{L}^{(t)} = \sum_{i=1}^{n} L(y_i, \hat{y}_i^{(t-1)} + f_t(x_i)) + \Omega(f_t). \tag{8.6}$$

We apply a second-order Taylor expansion around the previous prediction $\hat{y}_i^{(t-1)}$:

$$\mathcal{L}^{(t)} \approx \sum_{i=1}^{n} \left[L(y_i, \hat{y}_i^{(t-1)}) + g_i f_t(x_i) + \frac{1}{2} h_i f_t^2(x_i) \right] + \Omega(f_t). \tag{8.7}$$

Where:

- $g_i = \partial_{\hat{y}^{(t-1)}} L(y_i, \hat{y}_{i-1}^{(t-1)})$ is the Gradient.
- $h_i = \partial^2_{\hat{y}^{(t-1)}} L(y_i, \hat{y}_{i-1}^{(t-1)})$ is the Hessian.

Removing constant terms, the objective for the new tree f_t is:

$$\tilde{\mathcal{L}}^{(t)} = \sum_{i=1}^{n} \left[g_i f_t(x_i) + \frac{1}{2} h_i f_t^2(x_i) \right] + \gamma T + \frac{1}{2} \lambda \sum_{j=1}^{T} w_j^2. \tag{8.8}$$

A tree $f_t(x)$ can be defined as $w_{q(x)}$, where $q : \mathbb{R}^d \to \{1, ..., T\}$ maps a data point to a leaf index. We can rewrite the summation over instances i as a summation over leaves j:

$$\tilde{\mathcal{L}}^{(t)} = \sum_{j=1}^{T} \left[\left(\sum_{i \in_j} g_i \right) w_j + \frac{1}{2} \left(\sum_{i \in I_j} h_i + \lambda \right) w_j^2 \right] + \gamma T. \tag{8.9}$$

Let $G_j = \sum_{i \in I_j} g_i$ and $H_j = \sum_{i \in I_j} h_i$. This is now a collection of independent quadratic equations for each w_j.

For a fixed tree structure $q(x)$, we find the optimal weight w_j^* by setting the partial derivative $\frac{\partial \tilde{L}}{\partial w_j} = 0$:

$$G_j + (H_j + \lambda) w_j = 0 \Rightarrow \mathbf{w_j^*} = -\frac{\mathbf{G_j}}{\mathbf{H_j} + \lambda}. \tag{8.10}$$

Plugging w_j^* back into the objective function gives the **Structure Score** (or negative Quality Score) of the tree:

$$\tilde{L}^{(t)}(q) = -\frac{1}{2} \sum_{j=1}^{T} \frac{G_j^2}{H_j + \lambda} + \gamma T. \tag{8.11}$$

Since we cannot evaluate all possible tree structures, we use a greedy approach. For a potential split of a leaf into L and R children, the reduction in loss (the Gain) is:

$$Gain = \frac{1}{2} \left[\frac{G_L^2}{H_L + \lambda} + \frac{G_R^2}{H_R + \lambda} - \frac{(G_L + G_R)^2}{H_L + H_R + \lambda} \right] - \lambda \tag{8.12}$$